

1 Here are the first four terms of a sequence of fractions.

$$\frac{1}{1} \quad \frac{2}{3} \quad \frac{3}{5} \quad \frac{4}{7}$$

The numerators of the fractions form the sequence of whole numbers 1 2 3 4 ...

The denominators of the fractions form the sequence of odd numbers 1 3 5 7 ...

Write down an expression, in terms of n , for the n th term of this sequence of fractions.

(Total for Question 13 is 2 marks)

2 Here are the first five terms of a sequence.

−1 0 3 8 15

Find an expression, in terms of n , for the n th term of this sequence.

(Total for Question 2 is 2 marks)

3 Here are the first six terms of a quadratic sequence.

−1 5 15 29 47 69

Find an expression, in terms of n , for the n th term of this sequence.

(Total for Question 3 is 3 marks)

4 Here are the first five terms of a quadratic sequence.

10 21 38 61 90

Find an expression, in terms of n , for the n th term of this sequence.

.....
(Total for Question 4 is 3 marks)

5 Here are the first five terms of a sequence.

4 11 22 37 56

Find an expression, in terms of n , for the n th term of this sequence.

(Total for Question 5 is 3 marks)

6 The n th term of a sequence is given by $an^2 + bn$ where a and b are integers.

The 2nd term of the sequence is -2

The 4th term of the sequence is 12

(a) Find the 6th term of the sequence.

.....
(4)

Here are the first five terms of a different quadratic sequence.

0 2 6 12 20

(b) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

(Total for Question 6 is 6 marks)

- 7 The 2nd term of a geometric sequence is $3 + 2\sqrt{2}$
The 3rd term of the sequence is $13 + 9\sqrt{2}$

Find the value of the common ratio of the sequence.

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

You must show all your working.

.....

(Total for Question 7 is 4 marks)

8 Here are the first five terms of a geometric sequence.

$$\sqrt{5} \quad 10 \quad 20\sqrt{5} \quad 200 \quad 400\sqrt{5}$$

(a) Work out the next term of the sequence.

.....
(2)

The 4th term of a different geometric sequence is $\frac{5\sqrt{2}}{4}$

The 6th term of this sequence is $\frac{5\sqrt{2}}{8}$

Given that the terms of this sequence are all positive,

(b) work out the first term of this sequence.

You must show all your working.

.....
(3)

(Total for Question 8 is 5 marks)

9 $(2x + 23)$, $(8x + 2)$ and $(20x - 52)$ are three consecutive terms of an arithmetic sequence.

Prove that the common difference of the sequence is 12

(Total for Question 9 is 4 marks)